SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 2 – Architecture Design Assignment

Course Code:	CSA10 / CSA1005	**Course Title:**	Software Engineering
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – Architecture Design Assignment
Weightage:	20%	**Total Marks:**	25
CO2 Statement:	*Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills*		
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Date:		**Duration:**	60 minutes

Code of Conduct

I **KARNAM HARSHITH 192511384** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: _____

Annexure A – Architecture Design Assignment

Instructions to Students:

Each student will be assigned an **individual software project problem statement**. Based on the assigned problem, analyze the system requirements and design an appropriate software architecture by applying software engineering principles. Your submission should demonstrate your ability to select, justify, and evaluate a suitable architectural style.

Question

Based on your **assigned problem statement**, perform an **Architecture Design Analysis** and prepare an architecture design report by answering the following questions:

1. Analyze System Requirements

- Study the given problem statement and identify the system objectives.
- Analyze the functional and non-functional requirements that influence the software architecture.
- Identify key quality attributes such as scalability, maintainability, reliability, availability, performance, and security.

2. Select a Suitable Software Architecture

- Select an appropriate architectural style for the proposed system (e.g., Layered Architecture, Client-Server, MVC, Microservices, Event-Driven, Service-Oriented Architecture, Serverless, or Hybrid Architecture).
- Justify why the selected architecture is the most suitable for the given problem.

3. Draw the Software Architecture Diagram

- Design a clear architecture diagram illustrating the overall structure of the proposed system.
- Include all major layers/components, databases, external systems, APIs, and communication mechanisms.
- Use appropriate software architecture notations and labels.

4. Explain Components and Their Interactions

- Describe the purpose and responsibilities of each architectural component.
- Explain how the components communicate and exchange data.
- Illustrate the overall workflow of the system.

5. Discuss Quality Attributes

Evaluate how the proposed architecture addresses the following aspects:

- Scalability
- Security
- Performance
- Reliability
- Maintainability
- Availability

6. Justify Architectural Decisions

- Explain the rationale behind your architectural choices.
- Discuss the trade-offs considered while selecting the architecture.
- Explain how the chosen architecture supports future enhancements, system integration, and long-term maintainability.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
System Requirement Analysis (30%)	Thoroughly analyzes the problem statement, accurately identifies functional and non-functional requirements, and clearly explains quality attributes influencing the architecture.	Analyzes most requirements with minor omissions or limited explanation.	Identifies only basic requirements with partial analysis.	Requirement analysis is incomplete, inaccurate, or lacks clarity.
Selection of Software Architecture (25%)	Selects the most appropriate architectural style with strong technical justification aligned to system requirements.	Selects a suitable architecture with reasonable justification.	Architecture is partially suitable with limited justification.	Inappropriate architecture selected or justification is missing.
Architecture Diagram and Component Design (20%)	Produces a clear, complete, and well-structured architecture diagram with all major components,	Diagram is mostly complete with minor omissions or notation issues.	Diagram includes only essential components and lacks sufficient	Diagram is incomplete, unclear, or technically incorrect.

	interfaces, and interactions accurately represented.		detail.	
Component Interaction and Quality Attribute Analysis (15%)	Clearly explains component responsibilities, interactions, and thoroughly discusses scalability, security, performance, reliability, maintainability, and availability.	Explains most components and quality attributes with minor gaps.	Provides limited explanation of interactions and addresses only some quality attributes.	Component interactions and quality attributes are poorly explained or missing.
Architectural Justification and Technical Presentation (10%)	Provides strong justification for architectural decisions, discusses trade-offs and future enhancements, and presents a well-organized, professional report.	Provides adequate justification with minor gaps; report is well organized.	Limited justification with minimal discussion of trade-offs or future scope; report needs improvement.	Justification is weak or absent; report lacks organization and technical clarity.

Criteria	**Mark**
System Requirement Analysis **(30%)**	/5
Selection of Software Architecture **(25%)**	/5
Architecture Diagram and Component Design **(20%)**	/5
Component Interaction and Quality Attribute Analysis **(15%)**	/5
Architectural Justification and Technical Presentation **(10%)**	/5
TOTAL	/25

1. Study and Analyze the Scenario

1.1 Business Domain

The proposed system belongs to the IT service management and technical support domain. It is intended for organizations where employees depend on computers, applications, networks and other IT resources for their daily work. The platform acts as a common place for reporting, assigning, monitoring and closing IT service requests and incidents.

1.2 Stakeholders

1.3 Existing Challenges

IT problems may be reported through email or phone, making it difficult to keep all requests in one place. Tickets may not be prioritized consistently when the impact and urgency are not clearly recorded.

Support staff may spend time assigning tickets manually to the appropriate team. Employees may not have a clear view of the current status of their reported issue.

Common problems may be repeatedly reported because useful solutions are not maintained in one knowledge base.

SLA deadlines can be missed when response and resolution times are not monitored systematically.

Management may have limited information about ticket volume, resolution time and recurring technical problems.

1.4 System Objectives

Create IT service tickets with relevant issue details.

Categorize technical problems so that tickets can be handled by the right support area. Prioritize incidents using the impact and urgency of the reported issue.

Assign tickets to suitable support teams or staff.

Allow employees and support staff to track ticket status throughout the service lifecycle. Monitor service-level agreements and identify tickets approaching or exceeding SLA limits. Maintain a technical knowledge base for common problems and their solutions.

Collect user feedback after ticket resolution.

Generate basic IT service analytics for monitoring service performance.

1.5 Expected Outcomes

Faster resolution of IT issues. Improved service desk efficiency.

Better prioritization of important incidents.

Reduced repeated support requests through the knowledge base. Improved SLA compliance.

Better visibility into IT service performance.

2. Functional and Non-Functional Requirements

2.1 Functional Requirements

2.2 Non-Functional Requirements

3. Actors and Use Cases

3.1 Primary and Secondary Actors

3.2 Major Use Cases by Actor

3.3 UML Use Case Diagram

The diagram shows the main interactions between the four identified actors and the platform. The employee mainly initiates and follows service requests, support staff work on tickets, the IT manager monitors service performance, and the system administrator manages the supporting configuration and knowledge base.

4. Software Requirement Specification (SRS)

4.1 Introduction

The Intelligent IT Help Desk and Service Management Platform is proposed to organize the reporting and handling of IT-related problems in an organization. Instead of depending mainly on separate emails or phone calls, employees can use the platform to create a ticket and follow its progress. IT support staff can then manage the ticket from assignment to resolution, while managers can monitor service performance.

4.2 Purpose

The purpose of this SRS is to describe what the proposed platform should do, who will use it, the quality requirements it should satisfy, the information it should maintain and the limitations considered during requirement analysis.

4.3 Problem Description

In the current situation, technical issues can be reported through different channels. This can make ticket tracking, prioritization and assignment difficult. Employees may not know whether an issue has been received or when it is likely to be resolved. Support teams also need a consistent method for monitoring SLA targets and maintaining solutions for recurring problems. The proposed platform addresses these issues by providing a centralized ticket and service management process.

4.4 Scope of the System

The system covers IT incident and service-ticket reporting, categorization, prioritization, assignment, status tracking, SLA monitoring, knowledge-base management, feedback collection and service analytics. It is intended to support the internal IT help desk of an organization.

The system does not define the actual technical repair procedure for hardware or software. It records and manages the service process followed by the IT support team. Advanced AI-based diagnosis is also outside the minimum scope; the word 'intelligent' in this requirement analysis mainly refers to organized categorization, prioritization, assignment and service monitoring.

4.5 System Features

4.6 Use Case Descriptions

4.7 Data Requirements

4.8 External Interface Requirements

5. Assumptions and Constraints

5.1 Assumptions

Employees and support staff have access to the organization's internal network or approved internet connection.

Each user has a unique account and belongs to an appropriate role.

IT support teams and ticket categories are defined by the organization.

The organization decides the SLA targets for different types or priorities of tickets. Employees provide reasonably accurate information when creating tickets.

Knowledge-base articles are reviewed by authorized support staff before being treated as official solutions.

5.2 Constraints

6. Requirement Completeness and Consistency Validation

6.1 Stakeholder Coverage Check

The major stakeholders identified in the scenario are represented in the requirements. Employees have ticket creation, tracking and feedback requirements. IT support staff have assignment, status update and resolution requirements. IT managers have SLA monitoring and analytics requirements. System administrators have user, role, category, team and knowledge-base management responsibilities.

6.2 Completeness Check

6.3 Consistency and Ambiguity Check

The same ticket lifecycle terms are used consistently: Open, Assigned, In Progress, Resolved and Closed.

Employee and support-staff permissions are separated so that a normal employee cannot change another employee's ticket.

SLA monitoring is treated as a service-management requirement rather than a replacement for the actual technical repair process.

Priority is based on impact and urgency, which avoids using a single vague term such as 'important'.

Analytics are limited to service statistics described in the scenario, avoiding unsupported advanced prediction requirements.

6.4 Suggested Improvements

Define exact priority rules and SLA targets for the organization's actual service categories. Define escalation rules for tickets that remain unresolved close to the SLA deadline.

Add a standard ticket-category list so employees select from consistent options. Review knowledge-base articles periodically so outdated solutions are removed.

If required later, a future version could introduce automated text-based ticket classification and recommendation of knowledge articles.

7. Conclusion

The proposed Intelligent IT Help Desk and Service Management Platform provides a centralized way to manage internal IT service requests. The requirement analysis identifies the main stakeholders, functional and non-functional requirements, actors, use cases, data and interfaces. The SRS also considers assumptions, constraints and requirement validation. Based on the analysis, the proposed system can improve ticket visibility, prioritization, SLA monitoring and reuse of technical solutions while keeping the initial scope practical for an organizational help desk.

8. Requirement Traceability Summary

Stakeholder		Role / Interest in the System
FR-01	User Login	The system shall allow registered employees and support users to log in using valid credentials.
FR-02	Create Ticket	The system shall allow an employee to create an IT service ticket by entering the issue title, description, category and other relevant details.
FR-03	Ticket ID	The system shall generate a unique ticket ID for every newly created ticket.
FR-04	Categorize Ticket	The system shall classify a ticket into an appropriate technical category such as hardware, software, network or access.
FR-05	Prioritize Incident	The system shall assign a priority based on the reported impact and urgency of the issue.
FR-06	Assign Ticket	The system shall allow tickets to be assigned to a suitable IT support team or support staff member.
FR-07	Update Status	The system shall allow authorized support staff to update ticket status such as Open, Assigned, In Progress, Resolved and Closed.
FR-08	Track Ticket	The system shall allow employees to view the current status and basic progress of their own tickets.
FR-09	SLA Monitoring	The system shall record target response/resolution time and indicate tickets that are approaching or exceeding the defined SLA.
FR-10	Knowledge Base Search	The system shall allow users to search technical articles using keywords.

ID	Requirement	Description
FR-11	Knowledge Base Management	Authorized support users shall be able to add, edit and remove knowledge-base articles.
FR-12	Feedback	The system shall allow an employee to provide a rating and short feedback after a ticket is resolved.
FR-13	Analytics	The system shall provide IT service statistics such as ticket count, status distribution, priority distribution and resolution time.
FR-14	Notifications	The system shall notify relevant users when a ticket is assigned, updated, resolved or requires attention.
FR-15	Access Control	The system shall restrict functions according to the user's role.

ID	Quality Attribute	Requirement
NFR-01	Performance	Normal ticket pages and searches should respond within a few seconds under expected organizational usage.
NFR-02	Security	User authentication and role-based access shall protect ticket and administrative information.
NFR-03	Reliability	Ticket updates should be stored correctly without loss during normal operation.
NFR-04	Usability	The interface should use clear labels and simple forms so an employee can report an issue without special IT knowledge.
NFR-05	Scalability	The design should support an increase in employees, tickets and support teams without major redesign.
NFR-06	Maintainability	The system should use modular components so that categories, SLA rules and knowledge articles can be updated easily.
NFR-07	Availability	The platform should be available during normal organizational working hours with minimum planned downtime.
NFR-08	Auditability	Important ticket changes such as assignment, status and resolution updates should be traceable.

Actor	Type	Major Interaction
Employee	Primary	Creates and tracks tickets, searches knowledge base and submits feedback.
IT Support Staff	Primary	Handles assigned tickets, updates status, records resolutions and uses knowledge articles.
IT Manager	Primary	Monitors SLA performance, workload and service analytics.

Actor	Type	Major Interaction
System Administrator	Secondary	Manages users, roles, categories, teams and knowledge-base access.

Actor	Use Cases
Employee	Create Ticket; Track Ticket Status; Search Knowledge Base; Provide Feedback
IT Support Staff	View Assigned Tickets; Categorize/Prioritize Ticket; Update Ticket Status; Record Resolution; Search/Use Knowledge Base
IT Manager	Monitor SLA; View IT Analytics; Review Service Performance
System Administrator	Manage Users and Roles; Manage Categories and Teams; Manage Knowledge Base; Configure SLA Rules

Feature	Description
Ticket Management	Create, identify, assign, update and close service tickets.
Categorization and Prioritization	Classify issues and assign priority based on impact and urgency.
Service Lifecycle Tracking	Maintain the ticket status from creation until closure.
SLA Monitoring	Compare ticket response/resolution progress with defined service targets.
Knowledge Base	Store and search solutions for frequently occurring IT problems.
Feedback	Collect employee rating and comments after resolution.
Analytics	Present basic service desk statistics for management review.
Role-Based Access	Provide different permissions for employees, support staff, managers and administrators.

ID	Use Case	Actor	Precondition	Main Flow	Postcondition	Exception
UC-01	Create IT Ticket	Employee	Employee has a valid account.	Enter issue details → select/confirm category → submit ticket → system generates ticket ID.	Ticket is created with Open status.	Required issue information is missing.
UC-02	Handle Assigned Ticket	IT Support Staff	A ticket is assigned to the support staff member.	Open ticket → review details → diagnose → update status → enter resolution → mark Resolved.	Ticket contains updated status and resolution details.	Staff member is not authorized or ticket is unavailable.
UC-03	Track Ticket	Employee	Employee has created a ticket.	Open My Tickets → select ticket → view current status and updates.	Latest ticket status is displayed.	Ticket ID does not belong to the employee.

ID	Use Case	Actor	Precondition	Main Flow	Postcondition	Exception
UC-04	Monitor SLA	IT Manager	SLA rules have been configured.	Open SLA view → review tickets by deadline/status → identify overdue or at-risk tickets.	SLA performance information is displayed.	Required SLA data is unavailable.
UC-05	Search Knowledge Base	Employee / Support Staff	Knowledge articles are available.	Enter keyword → search → view matching article → use solution if relevant.	Relevant articles are displayed.	No matching article is found.
UC-06	Provide Feedback	Employee	Ticket has been resolved.	Open resolved ticket → give r ating/comment → submit feedback.	Feedback is stored against the ticket.	Ticket is not yet resolved.
UC-07	View IT Analytics	IT Manager	User has manager access and ticket data exists.	Open analytics → select perio d/category → view statistics.	Requested service statistics are displayed.	User lacks manager permission.

Data Entity	Important Data
User	User ID, name, email, department, role, account status
Ticket	Ticket ID, title, description, category, priority, status, created date/time, updated date/time
Assignment	Ticket ID, support team, assigned staff, assignment date/time
SLA	SLA type, target response time, target resolution time, current status, due time
Knowledge Article	Article ID, title, problem description, solution, category, last updated date
Feedback	Feedback ID, ticket ID, rating, comment, submitted date
Audit / Activity	Ticket ID, action, previous value, new value, user, date/time

Interface	Requirement
User Interface	Web-based interface with login, ticket form, ticket list, ticket details, knowledge-base search and dashboards.
Input Interface	Forms for ticket details, category/priority selection, status updates and feedback.
Output Interface	Ticket status screens, notifications, SLA indicators, knowledge articles and analytics summaries.
Database Interface	The application shall store user, ticket, SLA, knowledge-base and feedback information in a structured database.
Notification Interface	The platform may use email or in-application notifications for ticket updates.

Constraint Type	Constraint
Technical	The system depends on a web application, database and organizational network availability.

Constraint Type	Constraint
Operational	The quality of ticket categorization and assignment depends partly on the information entered and the configured rules.
Economic	The proposed system should be practical for an organization and should avoid unnecessary infrastructure or licensing costs.
Security / Legal	Access to employee and IT service information must be restricted to authorized users.
Time	This assessment focuses on requirement analysis and SRS preparation; full production implementation is outside the scope of this assessment.

Scenario Requirement	Status	Related Requirement
Ticket creation	Covered	FR-02, FR-03
Categorization	Covered	FR-04
Prioritization	Covered	FR-05
Assignment	Covered	FR-06
Status tracking	Covered	FR-07, FR-08
SLA monitoring	Covered	FR-09
Knowledge base	Covered	FR-10, FR-11
User feedback	Covered	FR-12
IT analytics	Covered	FR-13
Security / access	Covered	FR-01, FR-15 and NFR-02

Scenario Need / Problem	Requirement	System Response
Problem: scattered IT issue reporting	FR-02, FR-03, FR-08	Central ticket creation and tracking
Problem: difficult prioritization	FR-04, FR-05	Categorization and priority
Problem: manual assignment	FR-06	Support-team assignment
Problem: unclear service progress	FR-07, FR-08	Status lifecycle
Problem: missed SLA deadlines	FR-09	SLA monitoring
Problem: repeated common requests	FR-10, FR-11	Technical knowledge base
Problem: limited service visibility	FR-13	IT service analytics
Need for service quality feedback	FR-12	User feedback

UML Use Case Diagram

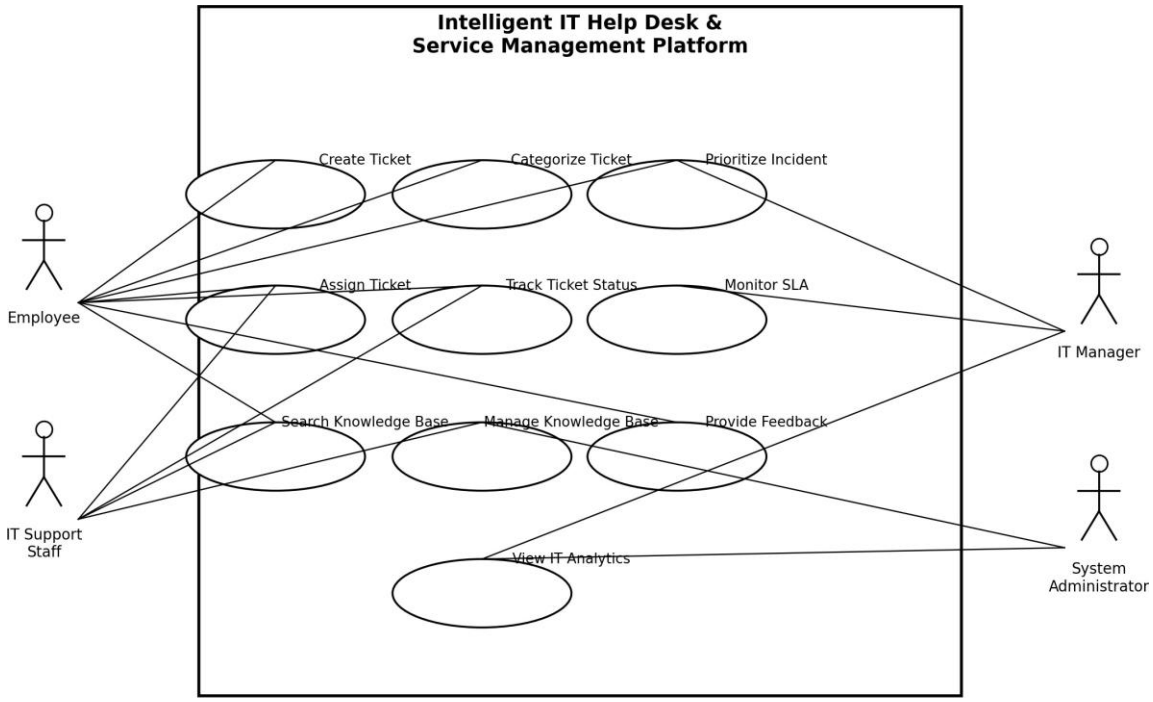


Figure 1. UML Use Case Diagram

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